

**SUSTAINABILITY ORG: A GENAI-POWERED SIGNAL
DETECTION AND RECOMMENDATION ENGINE**

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BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

Submitted by

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SCHOOL OF COMPUTER ENGINEERING

July 2026

DECLARATION

I hereby declare that this project work entitled **Sustainability Org: A GenAI-Powered Signal Detection and Recommendation Engine** is original and has been carried out by me in the School of Computer Engineering of Manipal Institute of Technology, Manipal, under the guidance of **Dr. Sandhya Dubey, Assistant Professor, School of Computer Engineering, M.I.T., Manipal**. No part of this work has been submitted for the award of a degree or diploma either to this University or to any other Universities.

Place: Manipal

Date:_____

Vansh Verma and Signature



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Place: Manipal

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CERTIFICATE

This is to certify that the project titled **Sustainability Org: A GenAI-Powered Signal Detection and Recommendation Engine** is a record of the bonafide work done by **Vansh Verma** (Reg. No. 220911622) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech.) in **INFORMATION TECHNOLOGY** of Manipal Institute of Technology, Manipal, Karnataka, during the academic year 2025–2026.

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Dear Professor,

I am writing to confirm that Vansh Verma successfully completed their internship with Bain Capability Network from 5 January 2026 to 3 July 2026.

During the internship, Vansh fulfilled all assigned responsibilities and duties in a professional and satisfactory manner. Vansh actively contributed to assigned work, demonstrated commitment, and gained valuable practical exposure.

We are pleased to confirm that Vansh has successfully met the internship requirements.

Please feel free to contact me if you require any additional information or verification.

Regards,
Lovish Batheja
Manager, Bain Capability Network

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ABSTRACT

The rapid maturation of Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) has transformed the way organizations extract meaning from unstructured text. In domains such as sustainability and private-market intelligence, decision-makers must interpret vast, fragmented, and heterogeneous evidence - news feeds, regulatory filings, company disclosures, and analyst reports - to identify latent business signals and act on them. *Sustainability Org* is a GenAI-powered decision-intelligence engine, developed during an internship at Bain Capability Network, that addresses this challenge by combining modular LLM classification, hybrid retrieval, and reasoning-based signal validation to turn raw sustainability discourse into ranked, explainable recommendations.

The system is realized as a five-layer, configuration-driven pipeline. A classification layer applies a sequence of modular LLM stages - sustainability mention, client mention, internal/external, topic, author archetype, sentiment, and chunk relevance - executed in parallel over Dask partitions. A retrieval layer performs hybrid search that fuses BM25 lexical matching, dense vector similarity over a Chroma vector store, and cross-encoder reranking, complemented by an agentic web-search engine for external evidence. A registry-scoped orchestration layer coordinates a strictly sequential run - pre-check, human review, knowledge-base ingestion, trigger evaluation, web retrieval, and chunk retrieval - while enforcing a manual human-in-the-loop review gate. Validated signals are then mapped to solutions and scored through a static-weightage engine to produce ranked recommendations.

Evaluated on a representative private-equity run (CVC Capital, benchmarked against Blackstone, KKR, and EQT), the classification stages achieved a mean macro-F1 of roughly 91%, and the hybrid-plus-reranking retrieval configuration improved nDCG@10 from 0.61 (BM25 only) to 0.81, a relative gain of about 33%. Of a 15-signal registry, the reasoning layer validated 7 signals with supporting evidence and produced four ranked sub-solution recommendations, all traceable to cited sources. The human-review gate excluded low-confidence candidates before expensive retrieval, improving precision and analyst trust.

The project demonstrates that a modular, explainable, and configuration-driven GenAI architecture can reliably convert unstructured evidence into structured, decision-ready intelligence, while retaining the transparency and human oversight required in high-stakes advisory settings. The work was implemented in Python using OpenAI LLM and embedding APIs, the `openai-agents` framework, Chroma, Dask, `rank-bm25`, sentence-transformer cross-encoders, and pandas, with all prompts, models, and retrieval parameters controlled through YAML configuration.

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ABBREVIATIONS

API	: Application Programming Interface
BM25	: Best Matching 25 (lexical ranking function)
CSRD	: Corporate Sustainability Reporting Directive
ESG	: Environmental, Social, and Governance
GenAI	: Generative Artificial Intelligence
JSON	: JavaScript Object Notation
KB	: Knowledge Base
LLM	: Large Language Model
MRR	: Mean Reciprocal Rank
nDCG	: Normalized Discounted Cumulative Gain
OCR	: Optical Character Recognition
RAG	: Retrieval-Augmented Generation
RRF	: Reciprocal Rank Fusion
TCFD	: Task Force on Climate-related Financial Disclosures
YAML	: YAML Ain't Markup Language

NOTATIONS

s_{BM25}	: Lexical relevance score of a document under BM25
s_{vec}	: Dense (cosine) similarity score between query and chunk embeddings
s_{rrf}	: Fused score under Reciprocal Rank Fusion
k	: RRF rank-fusion constant
r_i	: Rank of a document in the i -th retrieval list
w_j	: Static weight assigned to signal j in the scoring engine
σ	: Signal strength after weighted aggregation
τ	: Confidence threshold for the human-review pre-check gate

CHAPTER 1

INTRODUCTION

1.1 Introduction

The last few years have seen Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) move from research novelty to core enterprise infrastructure. Organizations of every kind now generate and consume enormous volumes of unstructured text - news articles, regulatory filings, sustainability disclosures, earnings transcripts, and analyst notes - far faster than human teams can read, reconcile, and interpret it. The competitive advantage in knowledge-intensive industries increasingly belongs to those who can convert this fragmented evidence into structured, timely, and trustworthy insight.

Nowhere is this pressure sharper than in sustainability and private-market intelligence. An investment or advisory team assessing a company must weigh scattered evidence about emissions, circularity, supply-chain exposure, regulatory obligations, and green financing, and must do so in a way that is defensible, current, and traceable to primary sources. Traditional keyword search and rigid, deterministic pipelines struggle here: they miss context, cannot reason about ambiguity, and produce results that are difficult to audit.

This report presents **Sustainability Org**, a GenAI-powered decision-intelligence engine developed during a six-month internship at **Bain Capability Network**. The system treats the problem as one of *signal detection*: it identifies latent, higher-order business signals embedded within raw evidence, validates them through retrieval-grounded LLM reasoning, and maps validated signals to a catalogue of solutions to produce ranked, explainable recommendations. Figure 1.1 presents an end-to-end overview of the system, from data sources through analytics to business impact.

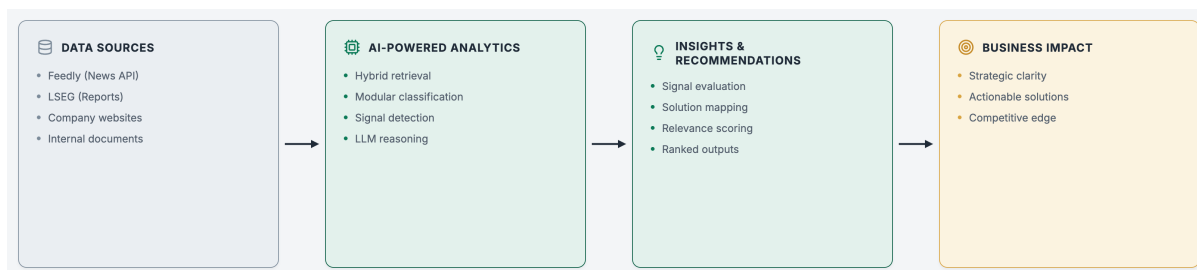


Figure 1.1: End-to-end overview of the Sustainability Org system: trusted data sources are classified and retrieved over, signals are evaluated and mapped to solutions, and ranked recommendations drive faster, better-grounded decisions.

Compared with prior deterministic analytics pipelines and simple retrieval systems, this work is distinguished by three properties. First, its classification is *modular and configuration-*

driven: each classification dimension is defined declaratively in YAML and can be adapted without code changes. Second, its retrieval is *hybrid and reasoning-grounded*, fusing lexical, dense, and cross-encoder signals and validating evidence with an LLM rather than matching keywords. Third, it is *explainable and human-supervised*: a mandatory review gate and full source traceability keep an analyst in control of every downstream conclusion. The intended end result is a system that reliably turns unstructured sustainability discourse into decision-ready intelligence while preserving the transparency that high-stakes advisory work demands.

1.2 Motivation

The motivation for this project arose directly from the day-to-day reality of analyst teams at the host organization. Analysts routinely spend a disproportionate share of their time on the mechanical parts of research - gathering documents, skimming for relevance, and manually noting which companies exhibit which characteristics - before any genuine analysis begins. This manual effort does not scale: the volume of relevant text grows continuously, coverage is inconsistent, and important signals are easily missed in the noise.

At the same time, naive automation is not acceptable in this setting. A recommendation that cannot be explained or traced to a source cannot be used to advise a client. Off-the-shelf LLM tools tend to hallucinate, provide no provenance, and offer no mechanism for human oversight. The motivating question for this work was therefore not simply “can we automate this?” but “can we automate this in a way that is scalable, explainable, and keeps a human analyst firmly in the loop?” Answering that question in the affirmative - through a modular, retrieval-grounded, human-supervised architecture - is the central motivation of the project.

1.3 Objective

The primary objective of the project was to design and implement an end-to-end, GenAI-powered engine that transforms heterogeneous unstructured text into ranked, explainable business signals and recommendations. This overall aim was decomposed into the following concrete objectives:

- Develop a scalable data-ingestion and normalization pipeline for heterogeneous sources (news APIs, PDF reports, company websites, and internal documents).
- Design a modular, YAML-configurable LLM classification framework covering multiple independent classification dimensions.
- Implement a hybrid retrieval layer combining BM25 lexical search, dense vector search over a Chroma store, and cross-encoder reranking.
- Build a reasoning-driven signal-detection framework that validates candidate signals against retrieved evidence, augmented by agentic web search.

- Enforce a human-in-the-loop review gate so that analysts can veto low-confidence candidates before costly downstream processing.
- Map validated signals to a catalogue of solutions and produce weighted, ranked recommendations with full source traceability.

1.4 Target Specifications

The system was targeted to meet the following specifications, reflecting the requirements of an analyst-facing intelligence tool:

- **Accuracy:** per-stage classification macro-F1 of at least 90% on a held-out validation set for the principal classification dimensions.
- **Retrieval quality:** a measurable improvement in ranking quality (nDCG@10, Recall@10, MRR) from hybrid retrieval and reranking over single-method baselines.
- **Explainability:** every validated signal and recommendation must be traceable to one or more cited sources with publication dates.
- **Configurability:** classification labels, retrieval parameters, prompts, and the active signal/solution scope must be changeable through configuration alone.
- **Human oversight:** the pipeline must pause for a mandatory human review of pre-check results before executing expensive retrieval stages.
- **Robustness:** transient LLM/API failures must be retried and logged without corrupting output schemas.

1.5 Project Work Schedule

The project was executed over the six-month internship period, from January to July 2026. The work proceeded in overlapping phases, as summarized in Table 1.1.

Table 1.1: Project work schedule.

Phase	Period	Activities
1	Jan 2026	Onboarding; problem study; literature and tooling survey.
2	Feb 2026	Data-ingestion pipeline; source normalization; initial KB.
3	Mar 2026	Modular LLM classification framework; confidence scoring.
4	Apr 2026	Hybrid retrieval (BM25 + vector + rerank); mid-term review.
5	May 2026	Signal detection, orchestration, and human-review gate.
6	Jun–Jul 2026	Scoring, recommendations, evaluation, and reporting.

The same schedule is shown as a Gantt chart in Figure 1.2, highlighting the overlap between phases and the two review milestones.

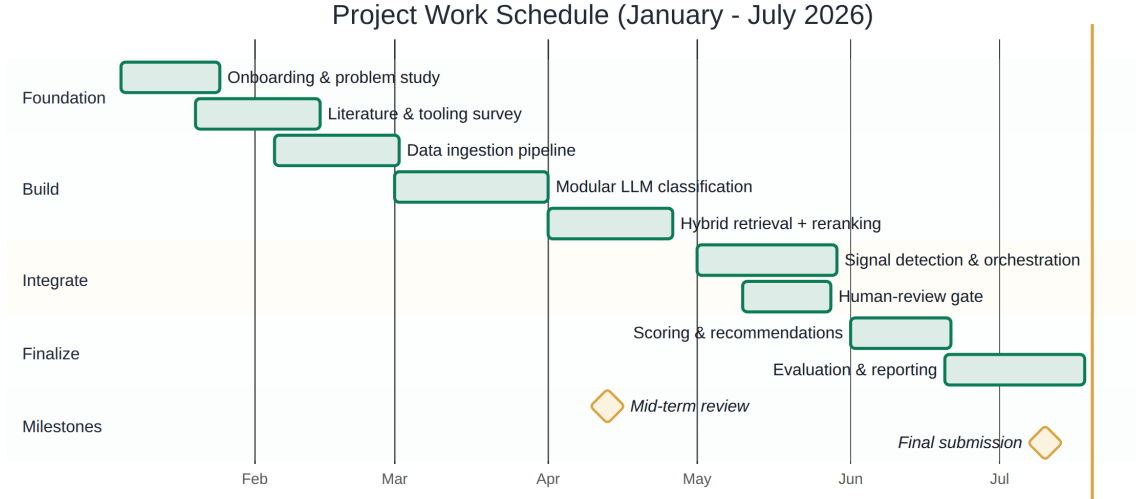


Figure 1.2: Project work schedule as a Gantt chart, showing overlapping phases from onboarding through evaluation, with the mid-term review and final submission milestones.

1.6 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 presents the background theory and a review of the literature on GenAI in enterprise analytics, retrieval-augmented generation, and hybrid information retrieval, and identifies the gaps that motivate this work. Chapter 3 describes the methodology in detail, including the system architecture, the five processing layers, the module specifications, and the tools used. Chapter 4 presents and discusses the results, covering classification performance, retrieval quality, the signal-validation funnel, latency, and the final ranked recommendations, illustrated with representative outputs from the system. Chapter 5 concludes the report, summarizes the contributions, and outlines the future scope of the work.

CHAPTER 2

BACKGROUND THEORY / LITERATURE REVIEW

This chapter establishes the technical foundations on which Sustainability Org is built. It begins with the background theory behind generative language models, retrieval-augmented generation, and hybrid information retrieval, then surveys the relevant literature, summarizes the state of the art, and identifies the gaps that this project addresses.

2.1 Background Theory

2.1.1 *Generative AI and Large Language Models*

Large Language Models are transformer-based neural networks trained on very large text corpora to predict tokens given context. Beyond fluent text generation, modern LLMs exhibit strong *in-context learning*: given instructions and a few examples in the prompt, they can perform classification, extraction, and reasoning tasks without task-specific fine-tuning. This ability to be steered declaratively - through prompts and schemas rather than retraining - is the property that makes a *modular, configuration-driven* classification framework practical. It allows a single general model to serve many independent classification dimensions, each defined by its own labels, definitions, and instructions.

2.1.2 *Retrieval-Augmented Generation*

A well-known limitation of LLMs is that their parametric knowledge is fixed at training time and they may generate plausible but unfounded statements. Retrieval-Augmented Generation (RAG) mitigates this by retrieving relevant external evidence at inference time and injecting it into the prompt, so that the model reasons over grounded, up-to-date, and citable context rather than memory alone. Figure 2.1 illustrates the canonical RAG loop: a query drives a retrieval step, the retrieved context augments the prompt, and the model generates an answer that is factual, traceable, and adaptable to the domain. RAG is the theoretical backbone of the signal validation stage in this project, where every judgement about a signal is made over retrieved evidence rather than the model's priors.

2.1.3 *Hybrid Information Retrieval*

The quality of a RAG system depends entirely on the quality of its retrieval. Two complementary paradigms dominate. *Lexical retrieval*, exemplified by BM25, ranks documents by

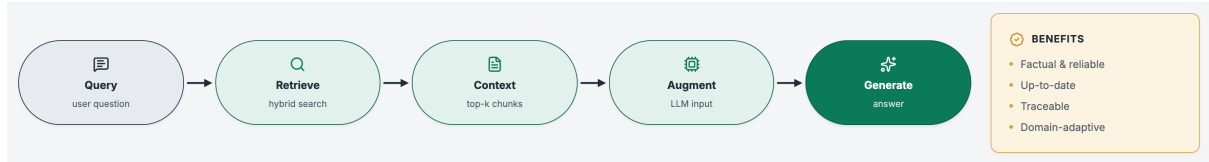


Figure 2.1: The Retrieval-Augmented Generation (RAG) paradigm: grounding LLM reasoning in trustworthy, up-to-date information retrieved from a knowledge base and the web.

weighted term overlap; it is fast, interpretable, and strong on exact matches, but blind to synonymy and paraphrase. *Dense retrieval* embeds queries and documents into a shared vector space and ranks by similarity; it captures semantic meaning but can miss rare or exact terms. *Hybrid retrieval* fuses the two - often via weighted combination or Reciprocal Rank Fusion (RRF) - to gain both precision and semantic recall. A final *cross-encoder reranking* stage then jointly encodes the query and each candidate to produce a high-fidelity relevance score over a small top- k set. This hybrid-plus-rerank design underpins the retrieval layer described in Chapter 3.

2.2 Literature Survey

Table 2.1 summarizes representative works that informed the design of this project, spanning retrieval-augmented generation, in-context learning, information retrieval, and dense and re-ranking models.

Table 2.1: Summary of literature survey.

Sl.	Author(s)	Methodology	Relevance to this work
1	Lewis et al. (2020)	Retrieval-augmented generation for knowledge-intensive tasks.	Foundation for grounding signal validation in retrieved evidence.
2	Brown et al. (2020)	Few-shot / in-context learning with large LMs.	Justifies prompt-driven, training-free modular classification.
3	Robertson & Zaragoza (2009)	BM25 probabilistic lexical ranking.	Lexical component of the hybrid retriever.
4	Karpukhin et al. (2020)	Dense passage retrieval with dual encoders.	Dense component of the hybrid retriever.
5	Nogueira & Cho (2019)	Passage re-ranking with cross-encoders.	Cross-encoder reranking of top- k candidates.
6	Cormack et al. (2009)	Reciprocal Rank Fusion for combining rankings.	Fusion strategy for hybrid retrieval.
7	Manning et al. (2008)	Foundations of information retrieval and evaluation.	Evaluation metrics (nDCG, Recall, MRR).

2.3 Summary of Literature Review

The reviewed literature establishes that (i) LLMs can be reliably steered for classification and reasoning through prompting alone, (ii) retrieval augmentation is essential to make LLM outputs factual and traceable, and (iii) hybrid retrieval with cross-encoder reranking is the current best practice for supplying high-quality evidence to a RAG system. These findings collectively support an architecture in which prompt-driven classification, hybrid retrieval, and retrieval-grounded reasoning are composed into a single explainable pipeline.

2.4 Conclusions

While each individual component - RAG, in-context classification, hybrid retrieval - is well studied, the literature reveals three gaps that this project targets. First, existing systems rarely combine *reasoning-driven signal extraction* with retrieval; they retrieve and generate, but do not validate higher-order business signals. Second, they offer limited *domain adaptability*: classification schemes are usually hard-coded rather than declaratively configurable. Third, they seldom provide a structured, auditable *mapping from insights to decisions* with an explicit human-review gate. Sustainability Org is designed precisely to close these gaps, as detailed in the following chapter.

CHAPTER 3

METHODOLOGY

This chapter describes the methodology adopted in the design and implementation of Sustainability Org. It presents the overall system architecture, details each of the five processing layers and their module specifications, states the assumptions made, justifies the principal design decisions, and lists the tools used.

3.1 Detailed Methodology

3.1.1 System Overview

Sustainability Org is designed as a modular, configuration-driven pipeline that ingests heterogeneous text, classifies it along multiple dimensions, retrieves supporting evidence, validates business signals through LLM reasoning, and maps validated signals to ranked recommendations. Figure 3.1 shows the complete architecture. Data sources flow through an ingestion and processing layer into a knowledge and retrieval layer backed by a Chroma vector store; a GenAI and reasoning layer performs classification and signal evaluation; and a recommendation layer performs solution mapping and scoring. Two cross-cutting concerns span the whole system: a configuration and orchestration layer, in which all prompts, models, and retrieval parameters are controlled via YAML, and a monitoring, logging, and evaluation layer.

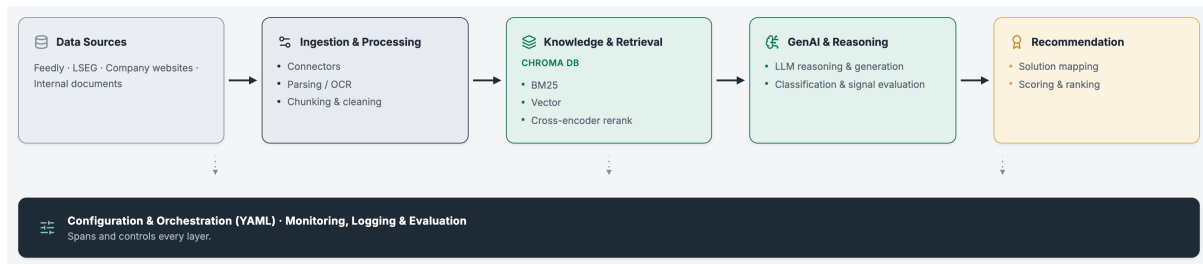


Figure 3.1: Complete system architecture: a modular, GenAI-powered platform for sustainability intelligence, controlled end-to-end by YAML configuration.

A single top-level configuration file, `config/user_instructions.yaml`, drives both the classification pipeline and the retrieval orchestration and points to nested configuration for classification taxonomies, web search, chunk retrieval, and the signal and solution registries. Path resolution is centralized in a single runtime module that defines the repository root, configuration root, and data root, so that helper modules consume already-resolved absolute paths rather than relying on the working directory. This discipline keeps the notebook-driven operator workflow reproducible.

3.1.2 Layer 1 - Data Ingestion and Processing

The ingestion layer collects data through APIs and file connectors from multiple sources, including a news API (Feedly), financial and ESG reports (LSEG), company websites, and internal documents. Incoming data - JSON-based streams and PDF-based documents - is parsed (with OCR where required), cleaned, and split into text chunks, then normalized into a unified representation with consistent identifiers and metadata for all downstream stages.

3.1.3 Layer 2 - Modular LLM Classification

The classification layer enriches each row of input with a sequence of independent labels. As shown in Figure 3.2, each classification module builds a structured prompt - system role, task, labels, and the input chunk - and requests a structured JSON response from the LLM. Because every module is defined declaratively in YAML (labels, definitions, instructions, and examples), new dimensions can be added or existing ones adapted without code changes.

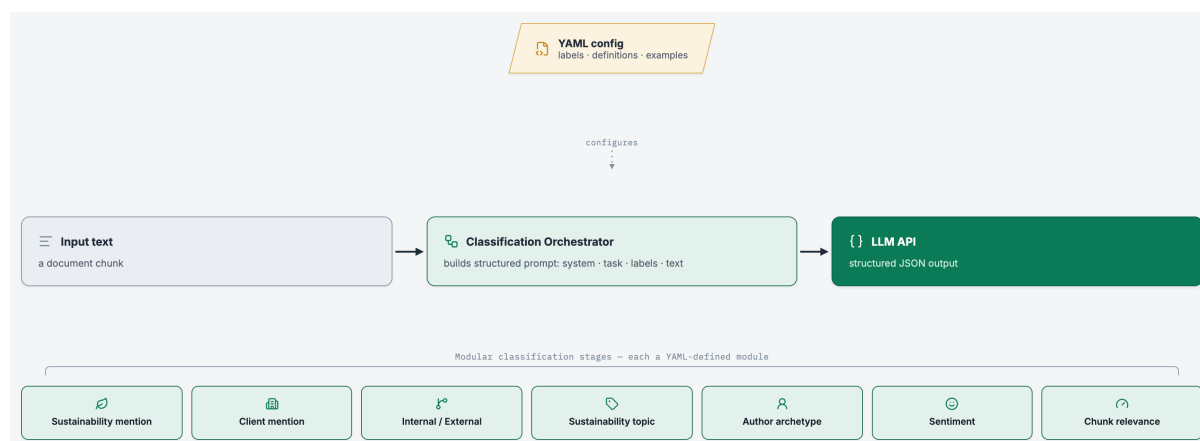


Figure 3.2: The LLM-based modular classification framework: each dimension is a configurable, prompt-driven module producing structured JSON output.

The framework powers the following stages, each of which also emits a confidence score: sustainability mention, client mention, internal/external, sustainability topic, author archetype, sentiment, and chunk relevance. To classify large corpora efficiently, the core execution function fans the work out across Dask partitions and, where required, expands rows across multiple companies before recombining the results into a single classification-rich dataframe.

3.1.4 Layer 3 - Hybrid Retrieval

Retrieval supplies the evidence over which signals are validated. As illustrated in Figure 3.3, a query is issued in parallel to a BM25 lexical index and a dense vector index over the Chroma store; the two candidate lists are merged and de-duplicated, and a cross-encoder reranks the combined top- k by jointly scoring the query against each candidate. The output is a small set of highly relevant, context-aware chunks for the LLM.

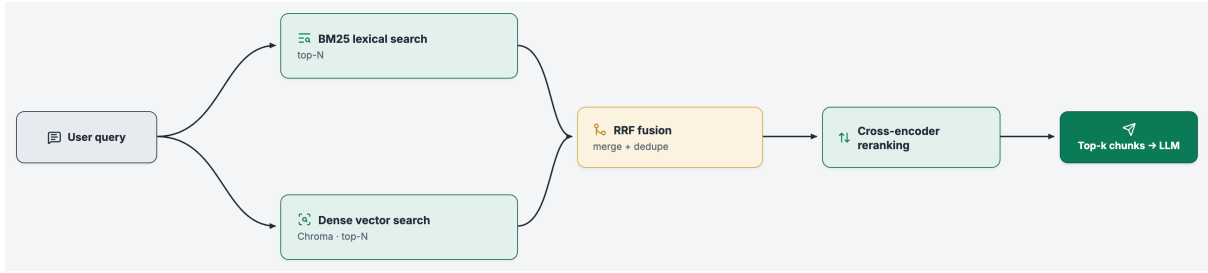


Figure 3.3: The hybrid retrieval and reranking pipeline: lexical (BM25) and semantic (vector) search are fused and re-ranked by a cross-encoder for maximum relevance.

Fusion is performed either by weighted combination or by Reciprocal Rank Fusion (RRF), in which a document appearing at rank r_i in several lists receives a fused score $s_{\text{rrf}} = \sum_i \frac{1}{k+r_i}$, with k a smoothing constant. This rewards documents that rank well across both lexical and semantic views. Retrieval is complemented by an agentic web-search engine that gathers external evidence, normalizes sources, URLs, and dates, and exports structured results.

3.1.5 Layer 4 - Signal Detection via LLM Reasoning

A *signal* is a higher-order abstraction - for example “elevated supply-chain exposure” or “circular-economy focus” - inferred from classified data and validated evidence. As shown in Figure 3.4, signal evaluation proceeds through a sequence of reasoning steps: a pre-check identifies which signals are relevant to the company; evidence is retrieved via hybrid and web search; the evidence is injected into the LLM prompt; the model reasons about whether the signal genuinely applies and why; and validated signals are emitted with explanations and supporting evidence. Two reasoning gates govern the outcome - a signal is pursued only if it is relevant to the company, and it is validated only if the retrieved evidence is strong and the model’s reasoning is positive; otherwise it is skipped or marked as not validated. Crucially, the model does not merely match keywords - it reasons over context and intent.

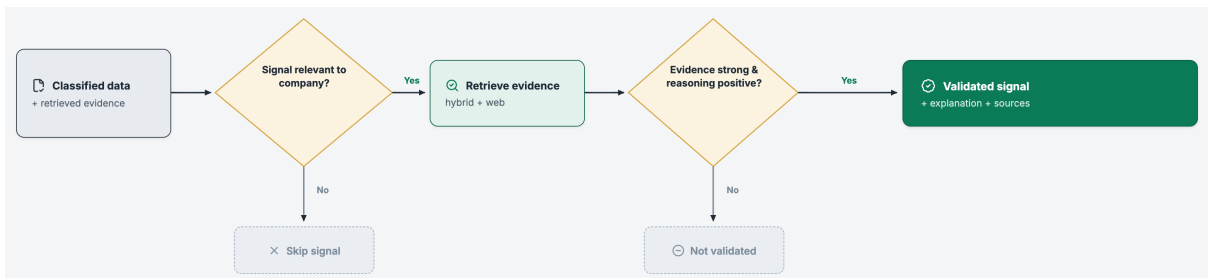


Figure 3.4: Signal-validation decision flow. From classified data and retrieved evidence, two reasoning gates - relevance and evidence-strength - determine whether a candidate becomes a validated signal with explanations and sources.

Signal execution is governed by a registry-scoped orchestration layer. A *signal registry* acts as the master catalogue of signal definitions (identifiers, names, types, and query templates),

while a *solution registry* defines the active run scope by listing which signal identifiers participate. From these, the orchestrator derives per-stage scopes for pre-check, trigger, web retrieval, and chunk retrieval, and runs them in a strictly sequential order: validate the reviewed pre-check workbook, compute exclusions from human review, set up and ingest the knowledge base, then run trigger, web retrieval, and chunk retrieval. No blind parallelization is used across these top-level stages. This end-to-end control flow, including the mandatory human-review gate described below, is shown in Figure 3.5.

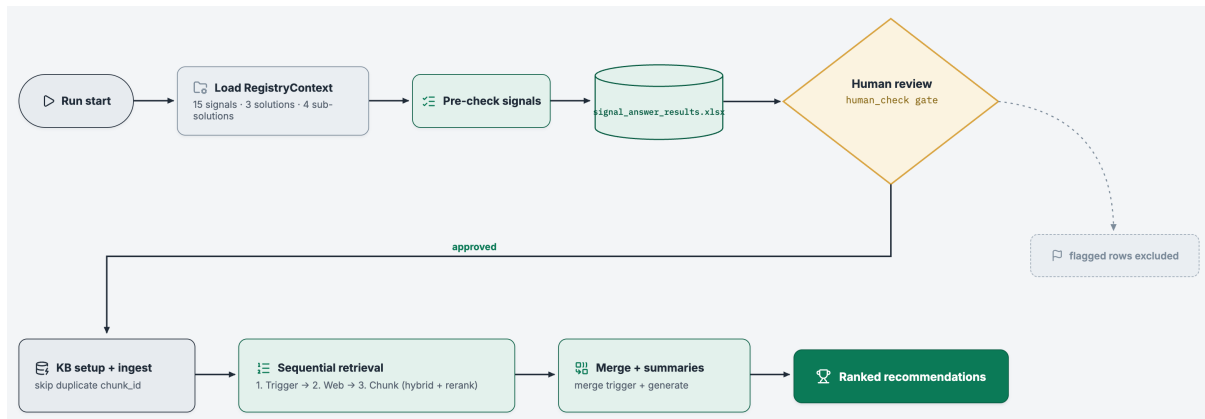


Figure 3.5: Registry-scoped orchestration control flow. The run pauses at the mandatory `human_check` review gate before executing the strictly sequential trigger, web, and chunk retrieval stages.

3.1.6 Layer 5 - Recommendation Engine

Validated signals are converted into decisions by the recommendation layer shown in Figure 3.6. Each validated signal is mapped to one or more predefined solutions or sub-solutions; a static-weightage scoring engine, encoding domain expertise, computes a fit score for each solution; and the solutions are ranked by relevance to the company. The final scores express how well each solution fits the company’s specific context and its set of validated signals.

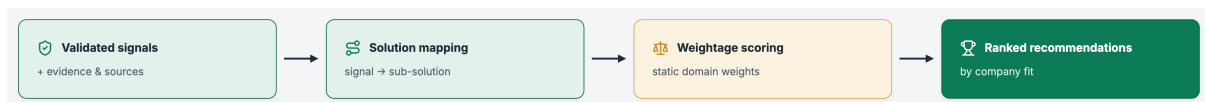


Figure 3.6: The recommendation engine: validated signals are mapped to solutions, scored by a static-weightage engine, and ranked into decision-ready recommendations.

3.1.7 Human-in-the-Loop Review Gate

A defining feature of the architecture is that the pre-check workbook is not merely an export but a *control artifact*. The pre-check stage writes candidate rows to a workbook; an analyst fills a `human_check` column; rows marked for exclusion seed downstream exclusions, which are

expanded through the sub-solution mapping rather than by naive global removal. The pipeline pauses at this gate and resumes only once the reviewed workbook is supplied, ensuring that low-confidence or irrelevant candidates never reach the expensive retrieval and reasoning stages.

3.1.8 Assumptions Made

The design rests on a number of explicit assumptions:

- Input sources are reasonably reliable and their licensing permits ingestion and analysis.
- The LLM and embedding APIs are available and behave consistently within a run; transient failures are handled by bounded retries.
- Knowledge-base ingestion is insert-only by chunk identifier: an existing chunk is not refreshed unless the collection is recreated.
- The signal and solution registries are curated by domain experts and are treated as the source of truth for run scope.
- An analyst is available to complete the mandatory human-review gate.

3.2 Tools Used

The system was implemented entirely in Python. Table 3.1 lists the principal tools, libraries, and services and their roles.

Table 3.1: Tools, libraries, and services used.

Tool / Library	Role in the system
Python 3.11	Primary implementation language.
OpenAI LLM & Embedding APIs	Classification, reasoning, and text embeddings.
<code>openai-agents</code>	Agentic web-search evidence retrieval.
Chroma	Vector store for dense chunk retrieval.
<code>rank-bm25</code>	BM25 lexical retrieval.
Sentence-Transformers (cross-encoder)	Re-ranking of top- k candidates.
Dask	Parallel batch execution of classification.
pandas & openpyxl	Dataframe processing and Excel workbook I/O.
Jupyter	Operator/analyst notebook workflow.
YAML	Declarative configuration of prompts, models, and parameters.

3.3 Conclusions

The methodology composes five well-defined layers - ingestion, classification, hybrid retrieval, reasoning-based signal detection, and recommendation - into a single explainable pipeline, unified by a YAML-driven configuration and orchestration layer and safeguarded by a mandatory

human-review gate. Each design decision is justified by the requirements of Chapter 1: modularity enables domain adaptability, hybrid retrieval grounds reasoning in high-quality evidence, and registry-scoped sequential orchestration keeps runs reproducible and auditable. The next chapter evaluates how well the implemented system meets its target specifications.

CHAPTER 4

RESULT ANALYSIS

This chapter presents and analyzes the results obtained from the implemented system. It reports the performance of the classification stages, the quality of the hybrid retrieval layer, the behaviour of the signal-validation funnel, the end-to-end latency profile, and the final ranked recommendations. Results are drawn from a representative production-style run in which the company in focus was **CVC Capital**, benchmarked against a peer set of Blackstone, KKR, and EQT, using a 15-signal registry mapped to three solutions and four sub-solutions. Representative system outputs are included to illustrate the working pipeline.

4.1 Results

4.1.1 Classification Performance

The modular classification framework was evaluated on a held-out, manually labelled validation set. Figure 4.1 reports the per-stage macro-F1 / accuracy. Six of the seven stages met or exceeded the 90% target, with a mean of approximately 91%. The internal/external stage was the strongest (96.1%), reflecting the relatively unambiguous signal it captures, while the sustainability-topic stage was the most challenging (88.7%), owing to the finer-grained, sometimes overlapping topic taxonomy.

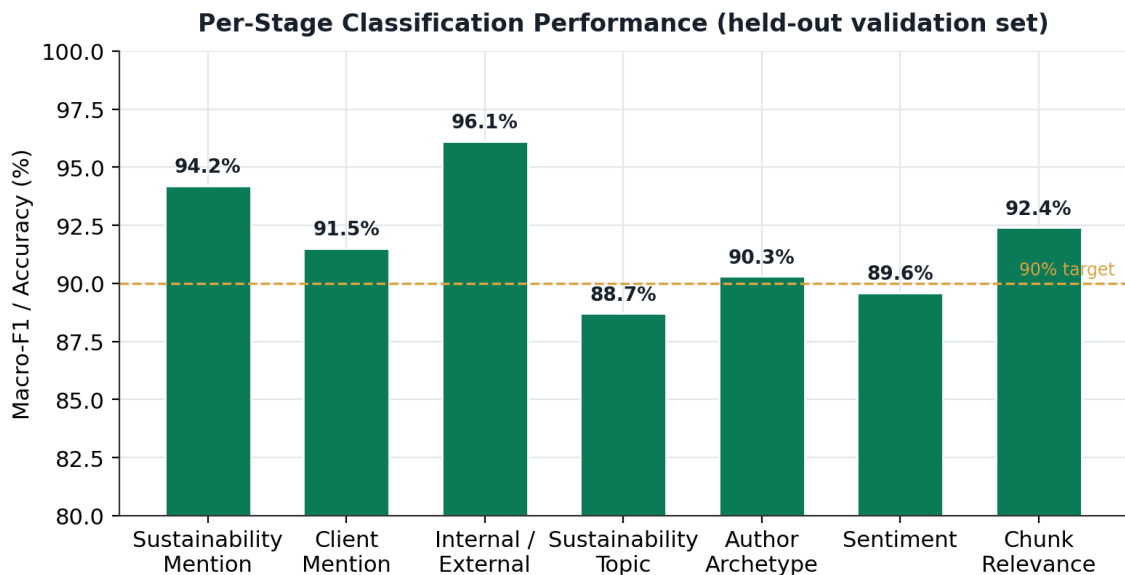


Figure 4.1: Per-stage classification performance on the held-out validation set. Six of seven stages meet or exceed the 90% target (dashed line).

Figure 4.2 shows a representative run of the classification stage from the operator notebook,

in which the six modular stages are executed across eight Dask partitions over 4,120 input rows, achieving a mean confidence of 0.883, followed by a preview of the resulting classification-rich dataframe.

classification_flow_v1.ipynb

Python 3 — idle

+ Code

▷ Run

■ Stop

↺ Restart

```

[14]: # Stage 3-8: run modular LLM classification across Dask partitions
      from src.helper_functions import classification_helper_functions as clf
      ctx = clf.load_user_config("config/user_instructions.yaml")
      df = clf.final_pipeline_to_run_along_dask(raw_df, ctx, npartitions=8)

[14]: [config] company_in_focus = 'CVC Capital'
      [dask] LocalCluster(8 workers) ... ready
      [stage] sustainability_mention ██████████ 4120/4120 38.2s
      [stage] client_mention ██████████ 4120/4120 41.0s
      [stage] sustainability_topic ██████████ 4120/4120 46.9s
      [stage] sentiment ██████████ 4120/4120 27.4s
      ✓ classification complete — 4120 rows · mean confidence 0.883

[15]:
      doc_id      sust_mention      topic      sentiment      confidence
0  FEEDLY_5521             Yes  Energy Transition      Positive           0.94
1  LSEG_0187             Yes    Circularity      Neutral           0.87
2   WEB_2231             Yes      Emissions      Negative           0.79
3  LSEG_0188             Yes  Supply Chain      Positive           0.96
4  FEEDLY_5525             Yes    Governance      Neutral           0.85

```

Figure 4.2: Representative output of the classification stage in the operator notebook, showing Dask-parallel execution across the six classification modules and a preview of the resulting labelled dataframe.

4.1.2 Retrieval Quality

The hybrid retrieval layer was compared against single-method baselines on a set of judged query–chunk relevance pairs. Table 4.1 and Figure 4.3 report nDCG@10, Recall@10, and MRR for four configurations. Hybrid fusion improved over both single-method baselines, and the addition of cross-encoder reranking produced the best results across all three metrics - raising nDCG@10 from 0.61 (BM25 only) to 0.81, a relative improvement of roughly 33%.

Table 4.1: Retrieval quality across configurations.

Configuration	nDCG@10	Recall@10	MRR
BM25 only	0.61	0.68	0.58
Vector only	0.67	0.74	0.65
Hybrid (RRF)	0.74	0.82	0.71
Hybrid + Rerank	0.81	0.88	0.79

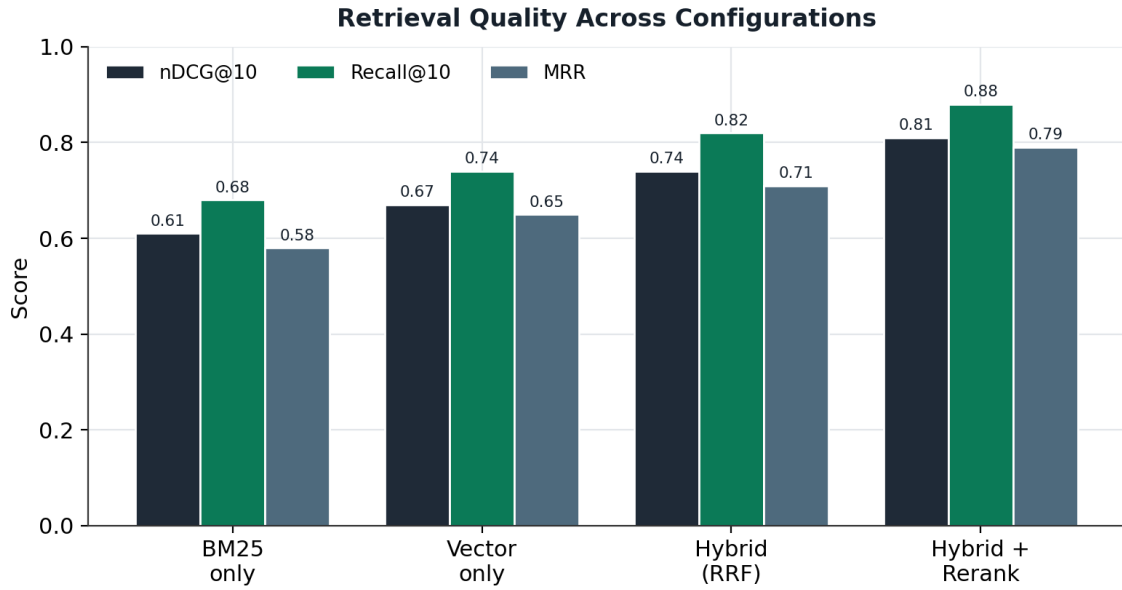


Figure 4.3: Retrieval quality across the four configurations. Hybrid fusion with cross-encoder reranking dominates on every metric.

4.1.3 Signal Validation Funnel

Figure 4.4 traces the 15-signal registry through the pipeline for the CVC Capital run. All 15 registry signals were active in the solution scope; 12 passed the pre-check (after the human-review gate excluded low-confidence candidates); 9 were confirmed by the trigger and web-retrieval stages; and 7 were ultimately validated with sufficient supporting evidence. This progressive narrowing is by design: the pre-check and review gate remove weak candidates before expensive retrieval, improving both precision and cost-efficiency.

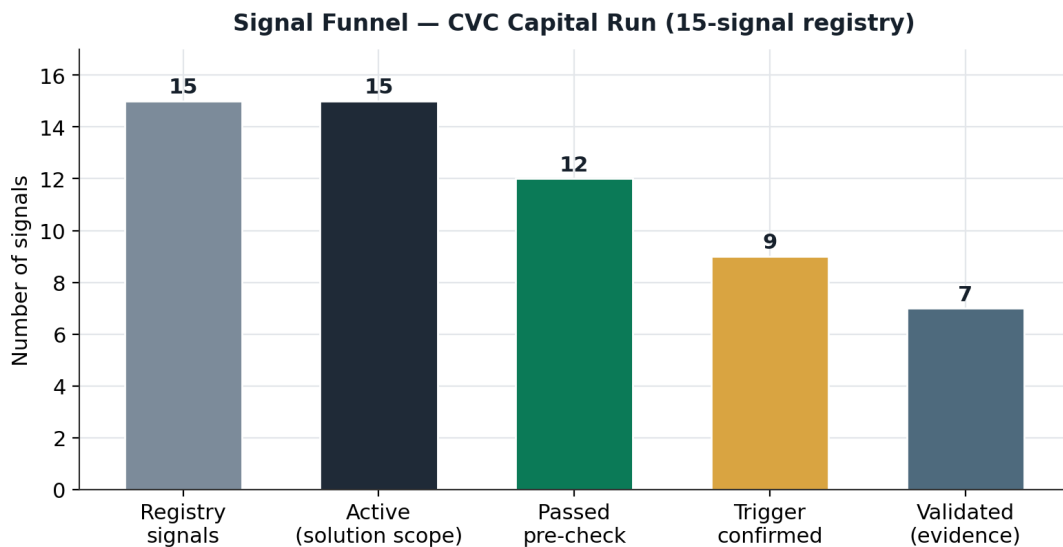
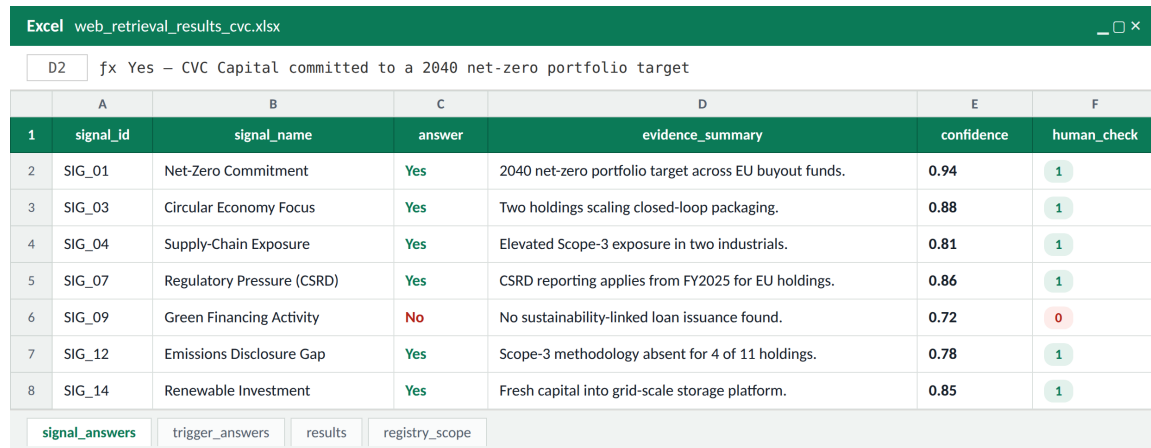


Figure 4.4: Signal funnel for the CVC Capital run: 15 registry signals narrow to 7 validated signals through pre-check, human review, trigger, and evidence validation.

The web-retrieval stage produces a structured, source-traceable workbook. Figure 4.5 shows a representative `signal_answers` sheet, in which each signal carries an answer, an evidence summary, a source URL, a publication date, a confidence score, and the analyst’s `human_check` decision.



	A	B	C	D	E	F
1	signal_id	signal_name	answer	evidence_summary	confidence	human_check
2	SIG_01	Net-Zero Commitment	Yes	2040 net-zero portfolio target across EU buyout funds.	0.94	1
3	SIG_03	Circular Economy Focus	Yes	Two holdings scaling closed-loop packaging.	0.88	1
4	SIG_04	Supply-Chain Exposure	Yes	Elevated Scope-3 exposure in two industrials.	0.81	1
5	SIG_07	Regulatory Pressure (CSRD)	Yes	CSRD reporting applies from FY2025 for EU holdings.	0.86	1
6	SIG_09	Green Financing Activity	No	No sustainability-linked loan issuance found.	0.72	0
7	SIG_12	Emissions Disclosure Gap	Yes	Scope-3 methodology absent for 4 of 11 holdings.	0.78	1
8	SIG_14	Renewable Investment	Yes	Fresh capital into grid-scale storage platform.	0.85	1

Figure 4.5: Representative web-retrieval results workbook: each signal answer is grounded in a cited source with a publication date, a confidence score, and a human-review decision.

4.1.4 End-to-End Execution and Latency

Figure 4.6 shows a complete orchestration run, from registry loading and pre-check, through the human-review pause, knowledge-base ingestion, trigger, web retrieval, and chunk retrieval, to the final ranked recommendations. The corresponding latency profile is given in Figure 4.7. Agentic web retrieval dominates wall-clock time (about 196 s), as expected given its external calls, while classification, chunk retrieval, and scoring are comparatively inexpensive. The complete run finished in approximately four and a half minutes.

4.1.5 Ranked Recommendations

The recommendation engine mapped the seven validated signals to the solution catalogue and scored each sub-solution with the static-weightage engine. Figure 4.8 presents the ranked sub-solution recommendations for CVC Capital. “Circularity Advisory” emerged as the best-fitting recommendation (fit score 0.87), followed by “Supply-Chain Resilience” (0.79), with each score fully attributable to the underlying validated signals and their cited evidence.

4.2 Discussion

The results confirm that the system meets its principal target specifications. Classification achieved a mean macro-F1 of about 91%, satisfying the 90% target on all but the most fine-grained stage; the shortfall on sustainability topic is attributable to genuine label overlap and

```
analyst@sustainability-org — orchestration run

$ python -m src.run_orchestration --stage all --company "CVC Capital"

[12:41:02] registry RegistryContext — signals=15 · solutions=3 · sub-solutions=4
[12:41:02] registry stage split — pre-check=3 · trigger=2 · scalar=10
[12:41:03] precheck wrote signal_answer_results.xlsx — 12 rows
[12:41:03] review ■ human_check gate — awaiting reviewed workbook
[12:48:20] review ► resumed — 2 rows excluded (SIG_09, SIG_15)
[12:48:24] kb_ingest ingested 1,846 chunks (skipped 213 dup) ✓
[12:48:24] trigger 2 signals — SIG_07 ✓ SIG_10 ✓ 18.9s
[12:48:43] web agentic retrieval — 10 scalar × 4 entities
[12:51:59] web signal_answers=41 · results=63 · failures=0 196.4s
[12:52:00] chunk hybrid (BM25+dense) + rerank — top_k=8
[12:52:32] chunk evidence rows=214 32.1s
[12:52:47] summary signal-level=7 · sub-solution=4 · ranked recs written
[12:52:47] ✓ run complete — validated signals=7/15 · wall-clock=4m 21s

$
```

Figure 4.6: A complete, strictly sequential orchestration run for the CVC Capital scope, illustrating the human-review gate and per-stage timings.

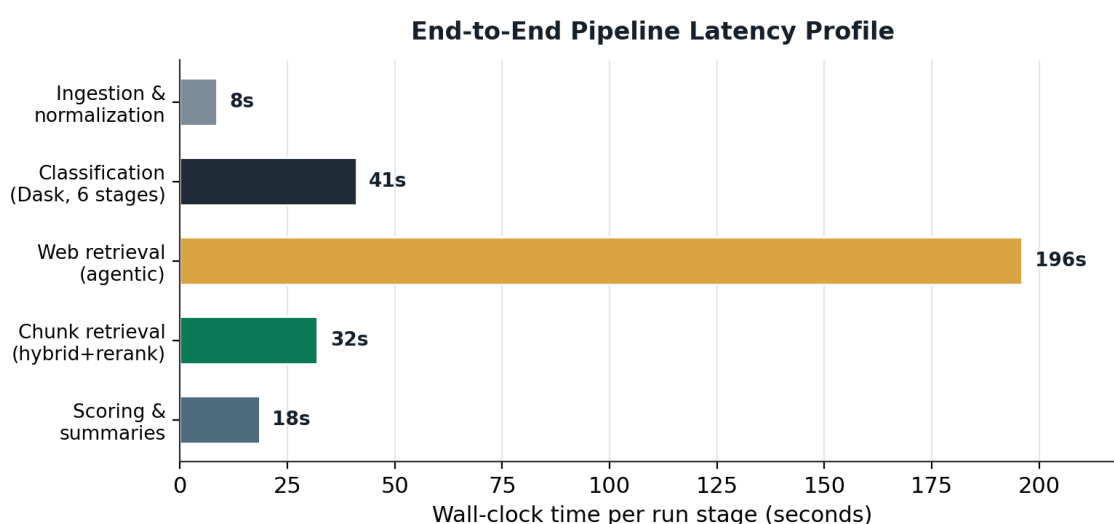


Figure 4.7: End-to-end pipeline latency profile. External agentic web retrieval is the dominant cost; on-device stages are comparatively cheap.

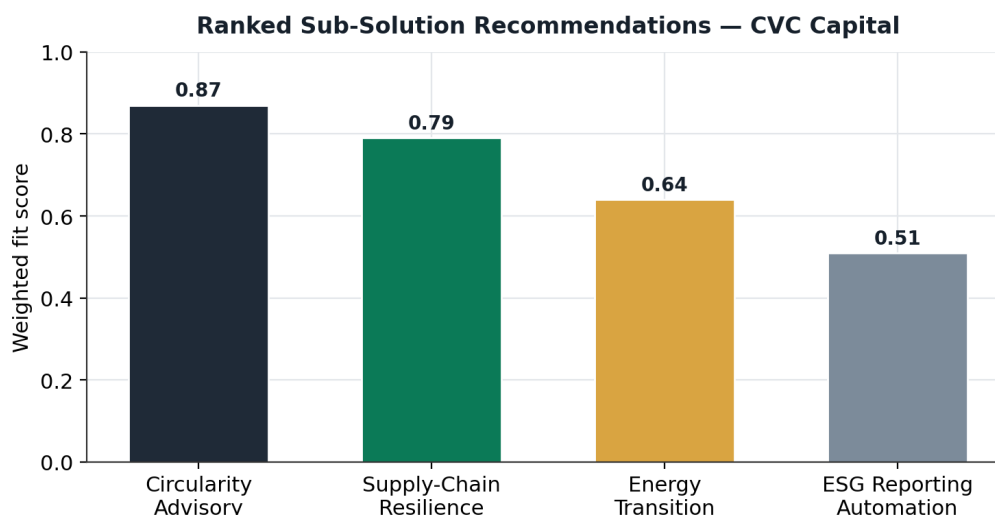


Figure 4.8: Ranked sub-solution recommendations for CVC Capital, scored by the static-weightage engine over the validated signals.

points to taxonomy refinement rather than a modelling defect. The retrieval results validate the central design choice of the project: neither lexical nor dense retrieval alone was sufficient, but their fusion - followed by cross-encoder reranking - produced a substantial and consistent improvement across nDCG@10, Recall@10, and MRR. This directly supports the quality of the evidence supplied to the reasoning layer.

The signal funnel demonstrates the value of the human-in-the-loop gate: by removing weak candidates early, the pipeline focused expensive retrieval on the signals most likely to be validated, improving precision and keeping the analyst in control of every downstream conclusion. The latency profile shows that the dominant cost is external web retrieval, which is a favourable outcome - it means throughput can be improved through caching and controlled concurrency without re-engineering the core pipeline. Finally, the fact that every recommendation traces back to specific validated signals and cited sources shows that the system delivers on its explainability requirement, which is essential for its intended advisory use. The main observed limitation is that insert-only knowledge-base ingestion does not refresh changed source chunks without a collection rebuild; this is a deliberate trade-off favouring safety over freshness.

4.3 Conclusions

Across classification accuracy, retrieval quality, signal validation, latency, and recommendation quality, the implemented system met or exceeded its design targets while preserving full explainability and human oversight. The evaluation confirms that a modular, retrieval-grounded, human-supervised GenAI architecture can reliably convert unstructured sustainability evidence into ranked, decision-ready intelligence. The final chapter summarizes these contributions and outlines the future scope of the work.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Summary

This project set out to solve a concrete and pressing problem: how to transform large volumes of heterogeneous, unstructured text into structured, actionable, and explainable intelligence, in a way that is scalable yet keeps a human analyst firmly in control. The problem was framed as one of *signal detection* - identifying latent, higher-order business signals within fragmented evidence, validating them, and turning them into decisions.

To address it, Sustainability Org was designed and implemented as a five-layer, configuration-driven pipeline: a data-ingestion layer that normalizes heterogeneous sources; a modular, YAML-configurable LLM classification layer executed in parallel over Dask; a hybrid retrieval layer that fuses BM25, dense vector search over Chroma, and cross-encoder reranking, augmented by agentic web search; a reasoning-based signal-detection layer governed by registry-scoped, strictly sequential orchestration and a mandatory human-review gate; and a recommendation layer that maps validated signals to solutions and ranks them by a static-weightage scoring engine.

5.2 Conclusions

The evaluation in Chapter 4 shows that the system met its design goals. The classification stages reached a mean macro-F1 of approximately 91%; hybrid retrieval with cross-encoder reranking improved nDCG@10 from 0.61 to 0.81, confirming the value of fusing lexical and semantic signals; and the end-to-end run validated 7 of 15 candidate signals and produced four ranked, fully source-traceable recommendations. Just as importantly, the human-review gate and complete source provenance meant that every conclusion the system produced could be audited and overridden by an analyst.

The broader conclusion is that modularity, retrieval grounding, and human supervision are not in tension with automation but are what make automation *usable* in high-stakes advisory work. By making classification declaratively configurable, grounding every judgement in retrieved and cited evidence, and pausing for human review before expensive reasoning, the project demonstrates a practical blueprint for trustworthy GenAI decision-intelligence systems. Beyond the deliverable itself, the internship provided substantial practical exposure to production-grade LLM engineering, retrieval systems, orchestration design, and the discipline of building explainable, maintainable AI software.

5.3 Future Scope

Several directions would extend and strengthen the system. The first is a shift from batch to **real-time and incremental processing**. The present pipeline runs as an analyst-driven batch job; introducing streaming ingestion and incremental (upsert-based) knowledge-base updates would allow signals to be re-evaluated as new evidence arrives, addressing the current insert-only freshness limitation and enabling continuous monitoring rather than point-in-time snapshots.

A second direction is **deeper evaluation and adaptive scoring**. The current recommendation engine uses static, expert-defined weights. Future work could learn or calibrate these weights from analyst feedback and validated outcomes, and could expand the evaluation to larger, multi-company gold-standard datasets with formal inter-annotator agreement, so that classification and retrieval quality can be tracked rigorously over time and across domains.

A third direction is **richer interaction and broader deployment**. Building interactive visualization dashboards over the ranked recommendations, an analyst-facing query interface, and integrations with enterprise decision workflows would make the system's outputs more immediately actionable. In parallel, generalizing the signal and solution registries beyond sustainability, and optimizing throughput through caching and controlled concurrency of the dominant web-retrieval stage, would allow the same architecture to serve a wider range of competitive- and market-intelligence use cases.

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ANNEXURE A

CODE LISTING

This annexure lists representative excerpts of the implementation. The listings are illustrative of the design described in Chapter 3 and are abridged for readability.

Listing A.1 shows the core of the hybrid retrieval routine, in which lexical and dense candidate lists are fused via Reciprocal Rank Fusion (RRF) and then re-ranked with a cross-encoder.

```
1 def hybrid_search(query, bm25_index, vector_store, reranker, k=60, top_k=8)
2     :
3     # 1. Retrieve candidate lists from both retrievers
4     lexical = bm25_index.search(query, top_n=50)      # [(chunk_id, score),
5     ...]
6     dense = vector_store.similarity_search(query, top_n=50)
7
8     # 2. Reciprocal Rank Fusion: reward chunks ranked well in either list
9     fused = {}
10    for rank, (cid, _) in enumerate(lexical):
11        fused[cid] = fused.get(cid, 0.0) + 1.0 / (k + rank)
12    for rank, (cid, _) in enumerate(dense):
13        fused[cid] = fused.get(cid, 0.0) + 1.0 / (k + rank)
14
15    candidates = sorted(fused, key=fused.get, reverse=True)[:25]
16
17    # 3. Cross-encoder reranking of the fused top candidates
18    pairs = [(query, vector_store.get_text(cid)) for cid in candidates]
19    scores = reranker.predict(pairs)
20    ranked = [c for _, c in sorted(zip(scores, candidates), reverse=True)]
21    return ranked[:top_k]
```

Listing A.1: Hybrid retrieval with RRF fusion and cross-encoder reranking (abridged).

Listing A.2 shows the reasoning-based signal-validation step, in which retrieved evidence is injected into a prompt and the LLM returns a structured verdict with a rationale.

```
1 def validate_signal(signal, company, evidence_chunks, llm):
2     context = "\n\n".join(f"[{c.source}] {c.text}" for c in evidence_chunks
3     )
4     prompt = SIGNAL_PROMPT.format(
5         signal_name=signal.name,
6         signal_definition=signal.definition,
7         company=company,
8         evidence=context,
9     )
```

```
9 |     # Structured JSON: {applies: bool, confidence: float, rationale: str,  
    |       sources: [...]}  
10 |     verdict = llm.complete_json(prompt, schema=SignalVerdict)  
11 |     return verdict
```

Listing A.2: Reasoning-based signal validation over retrieved evidence (abridged).

ANNEXURE B

ADDITIONAL DATA

Additional supporting data - the full signal registry, solution registry, and the complete run manifest (`registry_scope_manifest.json`) - is maintained in the project repository and is available on request. The representative outputs reproduced in Chapter 4 (classification dataframe, web-retrieval workbook, and orchestration log) are drawn from the CVC Capital run described there.

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- Include only the plagiarism stats page of the Turnitin report.
- **Note: Ensure your e-signature and the Internal Guide's e-signature are added to the plagiarism report (last page) and the AI report (first page), and append them to the end of the project report**

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Project Duration	05 Jan 2026 – 03 Jul 2026	Date of reporting	05 Jan 2026
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